

Enoch Zhu

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EDUCATION

The University of Texas at Austin

Austin, TX

B.S. Computer Science, Minor in Statistics and Data Science, Certificate in Japanese

Aug 2024 - May 2028

- Relevant Coursework: Operating Systems, Computer Architecture, Data Structures & Algorithms, Object-Oriented Programming, Linear Algebra, Statistical Machine Learning, Geospatial Artificial Intelligence

TECHNICAL SKILLS

- Infrastructure & DevOps: Linux, Docker, Nginx, Traefik, Cloudflare DNS, Tailscale, WireGuard, networking, CI/CD
- Programming Languages: Python, TypeScript, JavaScript, SQL, Bash, C, C++, Java, Go, Rust, Swift
- Backend & Data: PostgreSQL, PostGIS, pgvector, REST APIs, Supabase, FastAPI, Stripe Connect, distributed systems
- AI/ML: RAG, vector search, embeddings, query rewriting, tool-calling agents, MCP
- Certifications: CompTIA A+ (2023), GIAC Foundational Cybersecurity Technologies (2024)
- Languages: Mandarin Chinese (fluent), Japanese (intermediate)

EXPERIENCE

Icelis Global

Jun 2025 - Aug 2025

Software Engineering Intern

Doha, Qatar (Remote)

- Automated ESG report generation, loading document extractions into Postgres with per-job failure isolation
- Built a RAG chatbot over 1,076 pages of sustainability standards, using LLM query rewriting and page citations
- Ingested 40 GRI reporting standards into 2,081 chunks, using asymmetric embeddings and 429-aware retry backoff
- Shipped a time-tracking app for 50+ users with server-authoritative timestamps and a 4-role approval workflow

Sustainable Building Initiative

Aug 2024 - Aug 2026

Director of External Technologies

Austin, TX

- Led a 12-person engineering team across 7 repositories serving 75+ members and a \$200M+ project portfolio
- Built the member portal in Next.js and FastAPI, owning the Postgres schema behind auth, storage, and agent chat
- Proved zero cross-tenant reads over 21 tenant tables with adversarial tests from a 218-assertion pgTAP suite in CI
- Defended a link unfurler against SSRF and DNS rebinding by re-resolving hosts and re-validating up to 3 redirect hops
- Indexed a vector search that scanned every row, refitting 4096-dim embeddings under a 2000-dim index ceiling

Bell Flight

Aug 2023 - May 2024

Software Engineering Intern

Fort Worth, TX

- Cut reindexing time 90% on a legacy .NET and Apache Solr app, achieved by migrating to multi-threaded indexing
- Automated updates to 1,000+ customer access records with a Python script, eliminating manual record handling

PROJECTS

Private Cloud Infrastructure | Linux, Docker, Nginx, Cloudflare, Tailscale, WireGuard

Jun 2024 - Present

- Operated the SBI production backend for 75+ members and the Cuaya API, using self-hosted Dokploy on Linux
- Ran 20+ Docker containers and 7 health-checked services behind an Nginx reverse proxy with automated TLS
- Kept internal services at zero public ingress, reachable only over a private Tailscale and WireGuard overlay mesh
- Monitored 3 hosts and 20+ containers with Netdata, emailing alarms on disk, CPU, and memory pressure across 4TB+

Cuaya (Healthcare Provider Search) | TypeScript, Hono, PostgreSQL, PostGIS, Docker, Traefik

Jul 2026 - Present

- Cut provider search from 1s to 7ms over 2.9M US healthcare records, using PostGIS KNN with a query-plan test
- Streamed an 11 GB federal provider export into Postgres in idempotent 10k-row upserts, skipping unchanged rows
- Guarded irreversible deletes with a sweep that disarms itself on a 50% row-count drop until an admin re-arms it
- Drilled a full Postgres restore of a 397 MB dump to exact source row counts, plus a byte-exact R2-mirrored recovery
- Detected schema drift in CI by regenerating Drizzle migrations against a live PostGIS service in GitHub Actions

Bell Pepper Market | TypeScript, Supabase, PostgreSQL, PL/pgSQL, Stripe Connect, Expo

Sep 2025 - Present

- Built a farm-to-consumer delivery marketplace serving customers, farms, and drivers from one Postgres schema
- Prevented oversells at checkout across multiple farms, using row locks and server-side price re-reads per order
- Made farm payouts safe to retry after crashes, using a 15-minute lease and per-leg Stripe idempotency keys
- Scoped 9 Stripe webhook event types across 2 endpoints with separate signing secrets, rejecting mismatches with a 400

clickr (Open Source) | Rust, Linux, uinput, evdev, GitHub Actions, musl

Apr 2026 - Present

- Released an MIT-licensed Rust autoclicker for Wayland Linux, injecting clicks through a kernel virtual mouse
- Shipped static musl binaries for x86 and ARM from a tag-triggered pipeline with SHA-pinned third-party actions
- Minimized input-group privilege by refusing root, acting on only 7 hotkey codes, and releasing buttons on exit